Project Title: Active Directory Security Assessment

Project Description: The "Active Directory Security Assessment" project is a custom software tool that automatically scans a company's user and permission management system (Active Directory) to find security flaws. It doesn't require installing any extra software (agents) on corporate computers. It runs quietly in the background without slowing down daily business operations. It automatically detects bad configurations, hidden paths that hackers could use to steal admin privileges, old vulnerabilities, and deviations from global security standards. The tool only "reads" the data; it cannot modify or delete any system settings. It is powerful enough to handle massive corporate networks with over 4,000 users and complex rules. Instead of security experts spending days manually checking the system, this tool turns auditing into a fast, continuous, and automated process.

Suggested Method/Tool/Techniques(s) of Approach: We will write the codebase using advanced AI coding assistants like Claude Code or Antigravity. To prevent the AI from making things up (code hallucination) and to keep the code clean, we will use a step-by-step, module-by-module prompting strategy instead of generating everything at once. Architectural Pattern (Clean Architecture & Plugins): The software will follow Clean Architecture and SOLID principles. We are designing it as a plugin-based system. This means if we want to add a new security rule or compliance framework later, we won't have to touch the core engine or rebuild the entire project—we can just drop in a new isolated module (.dll or package). Technology Stack: We will develop the tool using high-performance, system-friendly languages like C# / .NET Core or Go. This allows the app to talk directly to Windows APIs and LDAP protocols without forcing the target machine to install any annoying third-party dependencies. Data Extraction (Agentless & Safe): LDAP Querying: To avoid crashing the network or running out of memory (RAM) when pulling thousands of user profiles, we will use smart, paginated queries to fetch data in small chunks. SYSVOL Parsing: We will directly read and analyze the Group Policy files from the Windows SYSVOL share to check password rules, user permissions, and basic registry settings. Analysis Engine: Once the data is collected, it will be processed in-memory by a detached Rule Engine. This engine will map the raw data against standard security interfaces (like IComplianceRule) to flag vulnerabilities. Future-Proofing (CI/CD Ready): The codebase will be structured so that it easily plugs into standard CI/CD pipelines. As new cyber threats emerge, we can automatically test and deploy new security rules without breaking the system.

Results And Deliverables Expected(from company's perspective): The primary deliverable of this internship project is a fully functioning, custom-built security tool. Instead of just using existing third-party products, developing this tool in-house provides significant strategic advantages and specific outcomes for the company. The expected results and benefits from the company’s perspective are organized below: 3.1. Deliverables and Technical Outcomes Actionable Security Intelligence: The tool will provide clear identification of immediate security risks (such as Kerberoasting vulnerabilities, stale privileged accounts, or weak access controls). These risks will be prioritized by criticality so the IT team knows exactly what to fix first. Automated Compliance Mapping: It will automatically map detected vulnerabilities to global security frameworks like ISO 27001. This drastically reduces the manual paperwork and workload usually required during major corporate compliance audits. Zero-Impact Execution: The tool guarantees a non-intrusive audit process. It operates strictly with read-only privileges and does not require "Domain Admin" rights for its primary scans. It performs zero write-operations to the Active Directory database, ensuring total business continuity without slowing down daily operations. Standardized Reporting: It will generate two types of structured outputs: Executive Reports (HTML/PDF): High-level security scores, charts, and visual risk indicators designed for management review. Technical Outputs (JSON/XML): Raw, structured data ready to be instantly plugged into the company's existing SIEM solutions or log management dashboards. 3.2. Strategic Justification: Why Build In-House instead of Using Free Tools? From a strategic standpoint, building this custom solution solves major limitations found in free, off-the-shelf tools: Data Privacy and Sovereignty (Zero-Trust): Third-party security tools handle the company's most sensitive data—its entire identity and access infrastructure. By developing a proprietary solution, the organization ensures that sensitive Active Directory data never leaves the local network, eliminating risks like data leaks, supply-chain vulnerabilities, or hidden telemetry from external freeware. Hyper-Customization for Corporate Infrastructure: Free tools offer generic, "one-size-fits-all" assessments. This custom engine allows the integration of specific corporate policies and tailored access controls designed for a large infrastructure with 4,000+ users. This creates a highly specific evidence-gathering mechanism that directly satisfies internal security committees. Full Automation without Paywalls: Many free or "community edition" tools block advanced automation, API access, or seamless SIEM logging behind expensive "Enterprise" paywalls. Developing this tool internally ensures it can be embedded directly into the company's existing CI/CD pipelines and automated workflows without facing licensing bottlenecks or vendor lock-in. Creating a Living Intellectual Asset: Instead of just consuming a temporary external product, the company will fully own a scalable asset. As the organization's network and hybrid cloud environments (like Microsoft 365) grow, internal teams can instantly adapt and update this tool without waiting for third-party software updates or paying recurring subscription fees.